

Developments of a Simple Model to Elucidate the Shape of Enveloped Viruses: Motivated by Monkeypox and SARS-CoV-2

Hua Deng

Advisor: Professor Chwen-Yang Shew

Doctoral Chemistry Program, Department of Chemistry

College of Staten Island, Staten Island, NY 10314

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Specific aim

First, we develop models and simulations that combine polymer and liquid-state physics to study how viral genome properties—such as shape, length, and flexibility—affect membrane morphology. By simulating genome behavior in confined spaces, we identify key physical principles governing viral assembly and structural stability.

Second, we study how internal pressure generated during genome packaging drives genome release into host cells. Understanding this process may reveal mechanisms of viral infection and inform the development of antiviral strategies, virus-inspired drug delivery systems, and improved vaccine design.

Motivation

In our previous work, we studied how spherical monomers assemble into dimers, trimers, and tetramers on curved surfaces, motivated by the trimeric spike proteins of SARS-CoV-2. Using Monte Carlo simulations, we modeled monomers that interact through directional (anisotropic) attraction and excluded-volume repulsion. We identified the interaction conditions under which trimer formation is energetically favored. The trimers formed in our simulations agree well with structural features observed in cryo-electron microscopy studies of SARS-CoV-2 spike proteins[1].

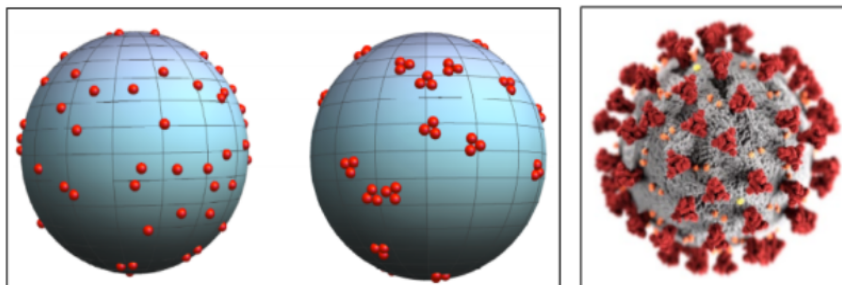


Figure 1: 1 Left image: Simulation of the SARS-CoV-2 spike protein on a spherical surface, illustrating the progression from randomly distributed monomers (Left) to trimer (Right) formations. Right image: SARS-CoV-2 for Covid-19 schematic image (CDC Public Health Image Library) [2][3]

Building on this work, we aim to develop a simple coarse-grained model with a minimal number of parameters to understand how interactions between the viral membrane and the enclosed genome control the shape of enveloped viruses. The goal is to gain thermodynamic insight into how genetic material is organized inside viruses, how the genome and membrane influence each other, and how internal pressure is generated by the packed genome. Understanding these physical mechanisms may help guide the development of new antiviral strategies.

Background and Significance

Viral infections significantly impact global health, driving pandemics and outbreaks. Enveloped viruses like Monkeypox and COVID-19 are surrounded by lipid membranes that protect their genomes and enable infection by interacting with host cell receptors. Understanding these mechanisms is key to developing effective treatments.

Virions are acellular particles lacking cellular structures such as organelles or membranes. Figure 2 illustrates the life cycle of an enveloped virus inside a host cell, from entry to the release of new virus particles. The virus first attaches to the cell surface and enters the cell. Once inside, it releases its genetic material into the cytoplasm. The host cell then reads this genetic information and quickly begins producing viral proteins. These early proteins help copy the viral genome and generate the components needed to build new viruses. Structural proteins are produced inside the cell and transported through the cell's internal membrane system, where they assemble together with the viral genome to form new virus particles. Finally, the newly formed viruses are packaged and released from the cell, allowing them to infect neighboring cells and continue the infection process.

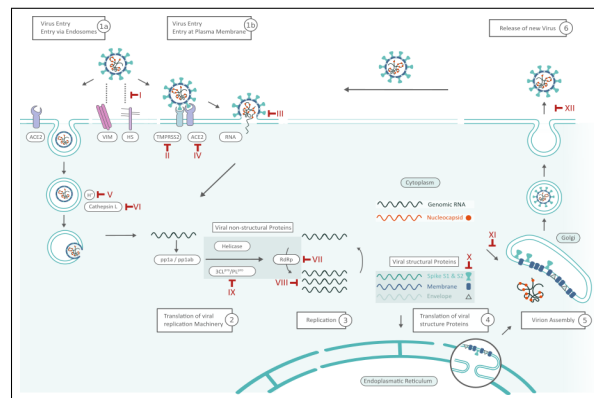


Figure 2: Life Cycle of an Enveloped Virus Inside a Host [4]

The shape of enveloped viruses is influenced by their genomes. Monkeypox virus has a ~ 190 kb

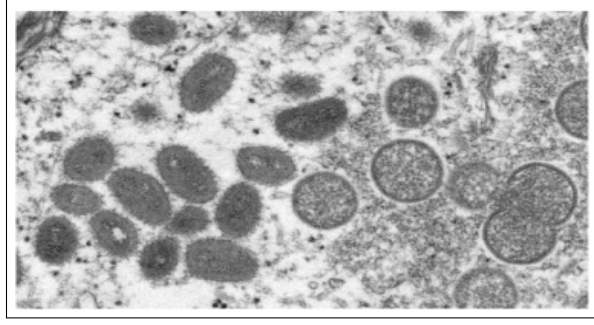


Figure 3: Electron microscopic (EM) image for Monkeypox virus particles. Oval-shaped virus particles are mature, and spherical particles are immature virions [2][3]

double-stranded DNA genome (~ 3000 nm contour length), much larger than the ~ 250 nm virus particle [5][6]. As shown in Figure 3, immature viruses are spherical, while mature forms appear oval.

This study investigates how genome shape influences membrane morphology in virus particles, focusing on the transition from spherical (immature) to oval (mature) forms. Using a coarse-grained model and Monte Carlo simulations, we will examine how genome compaction, fluctuations, and spatial arrangement drive membrane deformation during virus maturation.

In contrast, SARS-CoV-2 particles are smaller ($\sim 80 - 120$ nm) and spherical, with a single ~ 30 kb RNA genome ($\sim 1,400$ nm contour length) [7][8]. This difference suggests that genome length, stiffness, and organization play distinct roles in determining viral shape across different virus families.

Research Plan

I. Techniques

1. Coarse-Grained Models (CGM)

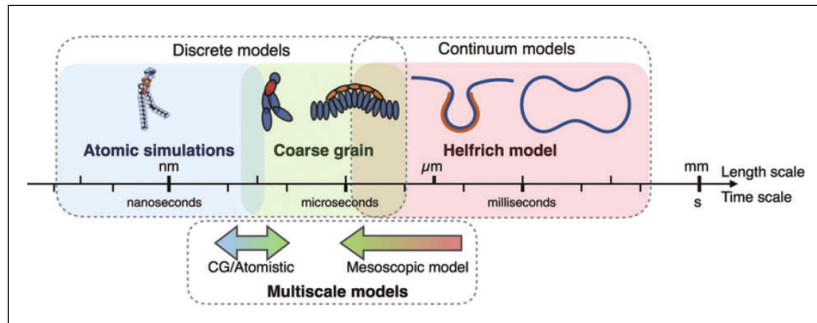


Figure 4: Different computational methods developed to study cellular membranes are valid in different length and time scales.[9]

It simplifies complex systems by grouping atoms or molecules into larger particles called beads. This enables large-scale simulations with reduced computational cost and identifying minimal-parameter

models.

2. Monte Carlo Simulation (MC)

Metropolis rule:

$$P = \min(1, e^{-\beta\Delta E}) \quad (1)$$

This is a key expression in the Metropolis-Hastings algorithm, often used in Monte Carlo simulations, to determine the acceptance probability of a proposed move in a system. P : Probability of accepting the proposed move; if accepted, positions update, otherwise the move is reverted. ΔE : Energy change from the move ($E_{\text{new}} - E_{\text{old}}$). β : Inverse temperature factor, $\beta = 1/(k_B T)$, from the Boltzmann distribution. If $\Delta E \leq 0$, the move is always accepted ($P = 1$) since it leads to a lower-energy, more favorable state.

II. Procedure and Methods

These are the specific models and procedures implemented using the above techniques.

1. Helfrich–Canham Membrane Model

The Helfrich membrane model is a standard physics model for describing how lipid membranes bend. The key idea is that bending a membrane costs energy, and the parameter κ (bending rigidity) controls how stiff the membrane is—larger κ means the membrane is harder to bend. The model also includes a spontaneous curvature C_0 , which represents a preferred natural curvature caused by effects such as protein binding or membrane asymmetry. Finally, we include constraints on the total surface area A and enclosed volume V , since real membranes cannot stretch freely and viruses maintain an approximately fixed size.

(1) Describes membrane bending energy:

$$F_{\text{full mem}} = \int_S \left[\underbrace{\frac{\kappa}{2} (2H - C_0(\vec{r}))^2}_{(1) \text{ spontaneous-curvature-modified bending}} + \underbrace{\bar{\kappa} K}_{(2) \text{ Gaussian-curvature term}} \right] dA + \underbrace{\lambda A}_{(3) \text{ area constraint}} + \underbrace{pV}_{(4) \text{ volume constraint}}. \quad (2)$$

(2) Discrete angle-based version used for simulations.

For simulations, the bending energy from the continuum membrane theory is rewritten in a more practical, discrete form using a triangulated surface. In this approach, the membrane is represented by connected triangular facets, and the local curvature is approximated by the dihedral angles between neighboring faces. This angle-based formulation is well suited for numerical implementation and enables efficient evaluation of membrane bending energy in Monte Carlo simulations, while still preserving the essential elastic behavior of the theoretical Helfrich–Canham model.

$$E_{\text{bend}} = \kappa \sum_{\langle i,j \rangle} (1 - \cos \theta_{ij}) + \lambda A + pV \quad (3)$$

2. Attration Force from Lennard–Jones potential

The coarse-grained model employs an effective interaction described by the Lennard–Jones potential. This potential incorporates a strong short-range repulsive term to prevent particle overlap and a weaker long-range attractive term that drives compaction and cohesive behavior.

$$U_{LJ}(r) = 4\varepsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad r \geq 2^{1/6}\sigma \quad (4)$$

σ is the distance at which the Lennard–Jones potential crosses zero, $U_{LJ}(r = \sigma) = 0$.

3. Excluded Volume Implementation

Excluded volume is essential in coarse-grained modeling to prevent particle overlap and maintain physical realism. It enforces the constraint that two particles cannot occupy the same space; without it, particles may overlap, interpenetrate one another, or unrealistically penetrate the membrane. To ensure that any two particles i and j do not overlap, we compute the squared center-to-center distance

$$d_{ij}^2 = (x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2. \quad (5)$$

For monodisperse particles of diameter σ , overlap is rejected by imposing the condition

$$d_{ij} < \sigma \quad (6)$$

To prevent particles from penetrating the membrane walls, the center of each particle is constrained

to remain at least a distance a (the particle radius) away from each face of a cubic simulation box of side length L , centered at the origin. This constraint is enforced by requiring

$$\begin{aligned} -\frac{L}{2} + r &\leq x \leq \frac{L}{2} - r, \\ -\frac{L}{2} + r &\leq y \leq \frac{L}{2} - r, \\ -\frac{L}{2} + r &\leq z \leq \frac{L}{2} - r. \end{aligned} \tag{7}$$

4. Simple Liquid Models

A simple liquid model treats the viral interior as a collection of coarse-grained particles and ignores detailed chemical structure. Instead, it focuses on the most important physical interactions, such as short-range attraction. In this framework, the total energy of the system is written as a sum of different contributions, including membrane bending energy and pairwise interaction energies between components such as genome segments, membrane particles, and crowding agents. Although this model is highly simplified, it captures the essential physics of a crowded viral environment and allows us to study key properties such as genome packing, internal pressure, and membrane deformation.

$$E = E_{\text{mem-bend}} + \sum_{\text{DNA-bonds}} U_{\text{LJ}} + \sum_{\text{DNA-mem}} U_{\text{LJ}} + \sum_{\text{DNA-crowder}} U_{\text{LJ}} + \sum_{\text{crowder-crowder}} U_{\text{LJ}} \tag{8}$$

5. Superellipsoid Model

A super-ellipsoid is a smooth 3D shape defined by the equation

$$\left(\frac{x}{a}\right)^n + \left(\frac{y}{b}\right)^n + \left(\frac{z}{c}\right)^n = 1 \tag{9}$$

We use the superellipsoid model because it provides a smooth and flexible description of virus geometry. The parameters a , b , and c determine the overall size and aspect ratios along the three principal axes, allowing the shape to be elongated, flattened, or brick-like. The shape parameter n controls the degree of surface roundness, tuning the transition from smoothly curved geometries to more box-like shapes with sharper edges (See Table 1). Together, these parameters enable realistic virus morphologies while maintaining well-defined curvature, which is essential for accurate calculation of membrane bending energy.

Table 1: Effect of the shape parameter n on superellipsoid geometry

n value	Shape behavior
$n = 2$	Ellipsoid (sphere if $a = b = c$)
$n > 2$	Rounded rectangular or brick-like shape
$n \rightarrow \infty$	Approaches a rectangular prism
$n < 2$	More pointed, diamond-like shape

The superellipsoid model provides a more physically realistic description of virus shape than a simple cubic model. Its smooth surface allows curvature to be well defined and enables continuous shape changes, which are essential for modeling membrane mechanics and shape evolution. A brief comparison between the cubic and superellipsoid shape models is summarized in Table 2.

Table 2: Comparison of Rectangular Prism and Superellipsoid Shape Models

Feature	Rectangular Prism Model	Superellipsoid Model
Realism	Sharp edges, less realistic	Smooth edges similar to real orthopox viruses
Curvature	No curvature; bending energy cannot be computed	Curvature varies spatially
Surface Area	Easy to compute analytically	Requires numerical integration
Shape Flexibility	Only cuboid shapes possible	Can smoothly interpolate between sphere, ellipsoid, and rounded rectangular shapes
Shape Transitions	It can only study cube to rectangle transition	Allows gradual and continuous shape transitions
Cost	Low computational cost	Higher computational cost
Overall Suitability	Less realistic virus modeling	More realistic and flexible for virus shape modeling

6. Ensemble Settings

(1) NVT (Canonical Ensemble):

In an NVT simulation, the number of beads, the volume, and the temperature are kept constant throughout the simulation. As the simulation progresses, the beads move and interact according to physical forces, leading to fluctuations in pressure and energy, even though the temperature remains stable.

$$P_{\text{accept}} = \min \left(1, \exp \left[-\frac{\Delta E}{k_B T} \right] \right) \quad (10)$$

In an NVT simulation, pressure fluctuates as beads move and interact. Plotting pressure versus Monte Carlo steps shows these changes. Total energy also fluctuates but remains near an average value. Plotting total energy over time helps check if the system is equilibrated.

(2) NPT (Isothermal–Isobaric Ensemble):

In an NPT simulation, the number of particles, pressure, and temperature are constant, while the volume fluctuates. As beads interact, the membrane can expand or contract to balance pressure, causing changes in density and structure. Bead positions, velocities, energy, and membrane size continuously evolve.

$$P_{\text{accept}} = \min \left(1, \exp \left[-\frac{\Delta E + p\Delta V - Nk_B T \ln \left(\frac{V_{\text{new}}}{V_{\text{old}}} \right)}{k_B T} \right] \right) \quad (11)$$

After obtaining simulation data, volume fluctuations at fixed pressures and temperatures will be analyzed. Average volume will be calculated and plotted against pressure. Additionally, volume and energy fluctuations over time will be examined to assess stability and equilibration, providing insight into the membrane's pressure response.

III. Hypothesis

Hypothesis 1:

Virus shape is determined by genome packing and by interactions between the genome, internal components, and the viral membrane.

Hypothesis 2:

During maturation, increased genome condensation raises internal pressure, which drives a transition from spherical to elongated virus shape.

IV. Expected outcomes

- (1) The shape of a virus and its genome architecture form simultaneously.

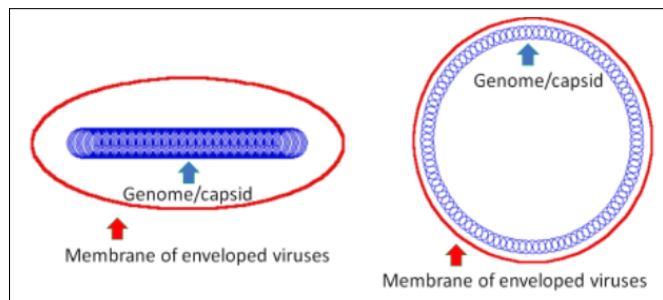


Figure 5: Preliminary 2D model studies to investigate the effect of the geometry of a genome on the shape of a virus. A rod-like genome induces an elliptic shape whereas a circular genome leads to a circular shape[2].

- (2) Starting from a cubic geometry, internal pressure from genome confinement combined with genome-crowder attraction breaks symmetry and drives elongation into an anisotropic shape.

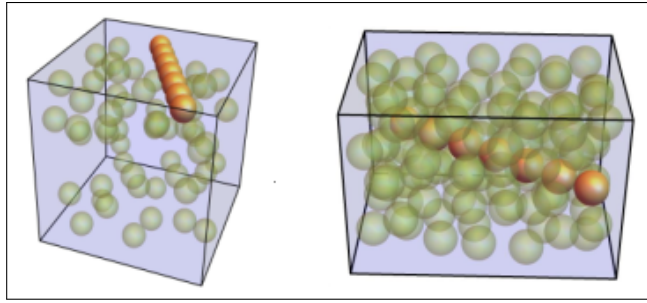


Figure 6: Initial cubic configuration (left) and final elongated cuboid configuration (right) obtained after sequential NVT equilibration followed by NPT simulation, consistent with the morphology of Monkeypox (Orthopoxvirus) particles.

V. Projected timeline

Goal 1: Build a Simple Virus Model

Build a simple model of a genome inside a soft membrane.

- Model the genome as connected beads.
- Model the membrane as a flexible surface.
- Include basic pushing (repulsion) and weak attraction.
- Study how the genome pushes on and deforms the membrane.

Goal 2: Study What Controls Virus Shape

Use the model to test what physical factors change virus shape.

- Change genome length (short vs long).
- Change genome stiffness (soft vs stiff).
- Change membrane size and flexibility.
- Measure changes in shape, pressure, and genome position.

Goal 3: Explain Shape Changes in Real Viruses

Use the model to explain experimental observations.

- Explain why COVID-19 particles stay mostly spherical.
- Explain why Monkeypox particles are elongated.
- Explain the change from spherical immature particles to elongated mature particles.

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